FLAT PULLBACK COMMUTES WITH SHEAF-LEVEL INTERSECTION PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0B0W: sources/stacks-chow/tags.txt, source lines 13874–13874. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AZF: sources/stacks-chow/tags.txt, source lines 13823–13823. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 079U: sources/stacks-chow/tags.txt, source lines 9284–9284. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a flat morphism of nonsingular varieties. Set $e = \dim(X) - \dim(Y)$. Let $\mathcal{F}$ and $\mathcal{G}$ be coherent sheaves on Y with $\dim(Supp(\mathcal{F})) \leq r$, $\dim(Supp(\mathcal{G})) \leq s$, and $\dim(Supp(\mathcal{F}) \cap Supp(\mathcal{G})) \leq r + s - \dim(Y)$. In this case the cycles $[f^*\mathcal{F}]_{r+e}$ and $[f^*\mathcal{G}]_{s+e}$ intersect properly and*

$$f^*([\mathcal{F}]_r \cdot [\mathcal{G}]_s) = [f^*\mathcal{F}]_{r+e} \cdot [f^*\mathcal{G}]_{s+e}$$

Proof. The statement that $[f^*\mathcal{F}]_{r+e}$ and $[f^*\mathcal{G}]_{s+e}$ intersect properly is immediate from the assumption that f has relative dimension e. By Lemmas 0B0W and 0AZF it suffices to show that

$$f^*\mathrm{Tor}_i^{\mathcal{O}_Y}(\mathcal{F}, \mathcal{G}) = \mathrm{Tor}_i^{\mathcal{O}_X}(f^*\mathcal{F}, f^*\mathcal{G})$$

as $\mathcal{O}_X$-modules. This follows from Cohomology, Lemma 079U and the fact that f^* is exact, so $Lf^*\mathcal{F} = f^*\mathcal{F}$ and similarly for $\mathcal{G}$. □

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